Mineral filler shall be non-plastic.

508.12.06 Blinding Sand - The Contractor shall be responsible for the supply of blinding material meeting the requirements of Table 508-D. That portion of blinding sand aggregate passing the 4.75 mm sieve shall be non-plastic.

TABLE 508-D GRADATION FOR BINDING SAND

SIEVE SIZE (mm)	PERCENTAGE PASSING
4.75	100
2.36	50 - 100
1.18	30 - 80
0.300	10 - 55
0.150	3 - 20
0.075	0 - 3

508.13 Graded Aggregate Seal Design - The Contractor shall provide a graded aggregate seal coat design, professionally prepared by qualified and experienced personnel. For this purpose, the Contractor shall use a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified registered member of the Applied Science Technologists and Technicians of British Columbia.

This design shall include the asphalt emulsion application rate in litres per square metre, spray temperature and the aggregate spread rate in kilograms per square metre. The design shall be presented to the Ministry Representative for review, at least three (3) days prior to the commencement of work and at any time subsequent changes are made to these rates.

The Contractor shall, upon request, submit to the Ministry Representative four litres of a representative sample of the emulsion(s) and a twenty kilogram sample of the aggregate(s) intended for use on the project. When requested, these samples shall be supplied at least three (3) days prior to the commencement of the work.

PART C – EQUIPMENT

508.20 General The Contractor shall supply plant and equipment necessary for the proper and efficient completion of the work, including clear and effective means of calibration.

508.20.01 The Ministry Representative shall have access at all times to any plant, equipment or machinery to be used on the Project.

508.20.02 If plant or equipment adjustment is necessary, it shall be done by a competent operator or mechanic or instrument repair technician, as required.

508.21 EQUIPMENT GUIDELINES – The following are guidelines and as such are not mandatory.

508.21.01 Pressure Distributor - The pressure distributor used for applying asphalt materials to consist of a fully insulated tank with a minimum capacity of 5000 ℓ, permanently and rigidly mounted on a truck or fifth wheel trailer provided with pneumatic tires and propelled by a power unit capable of maintaining a constant and uniform speed.

The pressure distributor should be provided with the following equipment:

a) A quick opening gate in the dome.

b) A dipstick calibrated in 20 mm increments and a calibration chart showing the volume for each 20 mm of depth, or provide the manufacturer dipstick and chart. Readily visible external depth gauge indicating the quantity of asphalt in the tank to the nearest 50 ℓ.

c) A means of heating the asphalt material uniformly to any temperature up to 125°C and maintaining the contents constantly at any selected temperature without any local overheating.

d) A heavy duty, industrial-type thermometer and a remote reading dial thermometer.

e) Rear-mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment. The spray bars to be capable of adjustment to provide a uniformly sprayed surface from a minimum of 0.6 m to at least 4 m width.

The spray bars to be capable of producing up to triple coverage at one pass. The feed to the spray bars must be arranged so that there is a constant circulation of hot asphalt material to the bars to ensure a uniform viscosity and constant pressure of the liquid asphalt at each nozzle both before and during spraying operations.

f) The feed manifolds to the spray bars to be provided with positive filtration to prevent clogging of bars and nozzles.

g) Rotary adjustable spray bar nozzles to be of the

same type, correct size and manufacture; and set to produce uniformly fan-shaped sprays without atomization. Nozzles to be inspected daily and any nozzle having nicked or damaged edges should be replaced.

h) Spray nozzles to be provided with valves capable of instant full opening and positive cut-off.

i) A connection to a circulating pressure manifold to which a hose may be attached for a single nozzle hand-operated outlet.

j) A pressure pump capable of applying the asphalt material at a minimum rate of 180 ℓ per minute, per metre of spray bar.

k) A pressure gauge accurate to within 15 kPa and a meter calibrated in litres per minute, should be readily visible.

l) A fifth wheel tachometer, calibrated in metres per minute and readily visible to the operator of the distributor.

m) Vehicle spring "tie-downs" to prevent the change in height of the spray bars due to the reduction of the asphalt load during spraying.

n) Electronic distance measuring device, with print out connected to the transmission, that is capable of measuring each spray length and the total length sprayed during the day.

508.21.02 Mechanical Spreader - Cover aggregate to be spread by means of an efficient, self-propelled spreader mounted on pneumatic tires and capable of continuously and uniformly spreading closely regulated quantities of crushed aggregates at the application rates selected. The loading hopper should be of such a capacity as to ensure continuous operation between dumps of aggregate. Aggregate to be transferred from the loading hopper to the front spreading device in such a manner as to ensure a uniform flow and prevent segregation of particles.

The spreading mechanism to be so designed as to distribute and spread aggregate across the selected width without segregation. Aggregate to be applied to the freshly sprayed surface by use of a comb screen for Class A and B Graded Aggregate Seals so that the largest particles are first placed on the road with the smaller fractions falling on top. The comb screen is to be mounted beneath the discharge openings of the spreader mechanism in such a manner that the angle of the comb screen relative to the road surface can be easily and quickly changed. Reverse flow aggregate spreaders are exempt from this clause.

Suitable provision to be made to prevent aggregate from rolling on the freshly sprayed surface. The cover aggregate should be applied ahead of the spreader wheels, with sharply defined, straight edges and without any lateral movement of the aggregate after initial application.

The spreader to be designed to apply aggregate at any application width between 2 m and 4.12 m. (Note: a 4.26 m hopper is required).

All joints should be at centreline or lane line.

508.21.03 Rollers - Self-propelled pneumatic tired rollers (minimum mass of 11,000 kg) to be used on the work.

All rollers shall be provided with equipment for spraying water continuously on all tires or wheels, while rolling is in progress, and also be provided with such means as may be required to keep the tires clean and free of adhering asphaltic binder.

508.21.04 Power Brooms - Brooming work to be carried out using rotary power brooms. The brooms to have sufficient width, power and brushing capacity to completely clean the surface of the standard lane to be treated or which has been treated within three coverages.

508.21.05 Water Truck - Water trucks to have a minimum capacity of 9000 ℓ. Water to be applied from a distributor of the pressure type, equipped with a spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water accurately and uniformly.

Splash plate type of distributors or those equipped with spray bars that eject fine streams of water will not be permitted.

The distributor to be provided with a satisfactory means for accurately measuring the quantity of water sprayed. Measuring equipment is to be calibrated. The Contractor shall make all necessary arrangements for obtaining water.

508.21.06 Tanks, Mobile Asphalt Heating, & Storage – Insulated liquid asphalt storage tanks should meet the following minimum criteria:

a) A heating system, capable to heat from 10° to 200°C.

b) An automated heating and control system with safety shutdown.

PART D – CONSTRUCTION

508.31 Daylight Hours Only – Construction shall be carried out during daylight hours only.

508.32 Traffic Control and Signing – The Contractor shall submit, for the approval of the Ministry Representative, a Traffic Control Plan at least seven days prior to the pre-construction meeting.

Traffic Control on Seal Coat projects shall be in accordance DBSS 194 as well as the Ministry's publication "Traffic Control for Work on Roadways".

The Traffic Control shall be maintained at all times when in the opinion of Quality Manager or the Ministry Representative, the risk of damage to the mat is likely to become a safety issue or to result in unnecessary vehicular damage from loosened rock. Additional traffic control to protect the quality of the product is also the responsibility of the Contractor.

Traffic through the work zone shall be limited to 50 km per hour, or as directed by the District Manager Transportation, until completion of final sweeping.

Extending traffic control beyond the end of the shift, in order to maintain safety and reduce vehicular damage risks, will not preclude the Contractor from providing and applying blinding material to further reduce these risks.

The Contractor shall, after discontinuing the application of product for the day, continue to maintain flag-person assisted traffic control over the extent of the day's production, for a minimum of 24 hours for work on numbered routes, 2 hours on other routes, or as specified in the Special Provisions.

508.33 Temperature and Weather – Air temperature shall be a minimum of 6°C for application of graded aggregate seals, and application shall cease when the temperature falls below that level.

No application shall take place when the weather is misty or rainy.

508.34 Grade Preparation – Immediately prior to the application of the graded aggregate seal coat on a previously constructed granular surface, the surface shall be watered, graded and compacted and/or broomed to provide a non-segregated, tightly knit surface free from loose rock. Cross-fall will be at 3% or as directed by the Designer and super-elevations developed uniformly and carried consistently through each curve.

508.35 Joints – All joints between adjacent passes of seal shall be at centreline or lane lines.

508.36 Communications – The flagpeople, pilot car or cars, sign truck, spreader and supervisor's vehicle shall have an effective electronic means of communication so that instructions and information can be quickly and accurately relayed over the total length of the control area.

508.37 Sweeping – Sweeping shall meet the requirements of DBSS 145, and DBSS 165 and the following:

a) Immediately prior to the commencement of the sealing operations, all loose aggregates, dust, dirt, caked clay or foreign materials shall be removed from the width of the surface to be treated by brushing with power brooms supplemented by hand push brooms, shovels or the use of a power grader, and where necessary by flushing.

b) Particular care shall be taken to thoroughly clean to outside edges of the strips to be treated and to ensure that the sweepings are not so deposited on the shoulder as to permit subsequent contamination of the treated surface.

508.38 Repair – Any patching or repair work required, in the opinion of the Designer, Quality Manager or Ministry Representative, before sealing operations begin, shall be done by the Contractor.

508.39 Double Seal, Second Pass – Where the design calls for a double seal, the second pass of the surface treatment shall commence not less than 24 hours after completion of the first pass, subject to acceptable weather conditions as detailed in DBSS 508.33, and as the Quality Manager may direct.

PART E - QUALITY MANAGEMENT

508.41 Quality Management by the Contractor – The contractor shall be responsible for providing all resources required to carry out the quality control and quality assurance on all the constituent materials, processes and products, their testing and inspection, within the construction of the seal coat and the quality of the end product. (See DBSS 508 Appendix E - Quality Control Requirements and Guidelines).

508.41.01 Quality Control Plan – The Contractor shall prepare and submit a Quality Control Plan, in accordance with the Contract requirements, at least 7 days prior to the pre-construction meeting.

508.41.02 Sampling, Testing and Observing – The Contractor shall provide properly maintained equipment, qualified personnel and shall maintain access to a qualified

laboratory so that quality control tests and observations are performed and the results must be reported to the Quality Manager and, when requested, to the Ministry Representative within 48 hours. For asphalt emulsions, the test results must be received within 21 days of sampling.

The Contractor's Quality Control Plan shall provide details of the equipment provided, its calibration, and application methodology to meet the Contract requirements.

508.42 Quality and Quantity Control Records

508.42.01 Emulsion Quality – The Contractor is referred to tests performed on asphalt emulsion (DBSS 508 Appendix A). Quality control tests results for each batch of emulsion supplied to the job site must be made available to the Ministry.

508.42.02 Aggregate Quality–

The Contractor shall assess the supplied aggregate to determine and satisfy themselves as to its suitability regarding quantity and quality for the intended application. Any modifications deemed necessary to ensure a quality product shall be made at the Contractor's expense.

508.42.03 Emulsion Usage – Daily emulsion usage shall be recorded on the Form "Emulsion Usage" (DBSS 508 Appendix C). Information shall be made available to the Ministry Representative within 48 hours.

508.42.04 Aggregate Usage – The Contractor shall monitor the aggregate spread rate on a production day basis and make the results available, in an acceptable format, to the Ministry Representative within 48 hours.

508.42.05 Daily Activity Log – The Contractor's Superintendent or designate shall complete and make available to the Ministry Representative the "Daily Activity Log" (DBSS 508 Appendix D) within 48 hours. This form shall be completed and submitted for all calendar days from the first day of mobilization to the last day of demobilization, for any day on which work was performed on Site.

508.42.06 Emulsion Sampling

i) The Contractor shall, procure and store a four litre sample of the emulsion from each tanker for use by the Ministry and/or Contractor.

These samples shall be obtained, handled and stored in accordance with ASTM procedure D 140 "Standard Practice for Sampling Bituminous Materials" and also with the manufacturer's recommendations.

ii) The samples shall be identified and retained by the Contractor until demobilization at which time permission may be granted by the Ministry Representative to discard the samples in an environmentally friendly manner.

iii) The Ministry Representative may provide the Contractor with a copy of the test results from the Ministry's Quality Management program. The Contractor shall not utilize these tests in place of their own Quality Management Program.

iv) The absence of Ministry test results shall not relieve the Contractor's obligation to remedy any defect in materials or construction.

508.42.07 Final Quantity Report – The Contractor shall supply to the Ministry Representative, prior to final acceptance of the work, the total amount of emulsion and graded aggregate seal aggregate used on the project.

PART F - END PRODUCT ASSESMENT

508.51 General - The Contractor shall provide an end product conforming in accuracy and detail as specified by the contract. Payment is based on end product assessment by the Ministry Representative.

508.52 Surface Deficiency Definitions – See DBSS 508 Appendix F "Surface Defects And Photographs" for examples and details of the conditions listed below. For the photographs of the various deficiencies, see SS 508 Appendix F in the *Standard Specifications for Highway Construction.*

508.52.01 Blackening – Blackening is the emergence of excess asphalt onto the finished surface whereby the finished surface becomes discoloured but the coarse surface texture remains. Under this condition, the effective cover stone can still be seen to protrude above the level of the asphalt binder's upward migration

508.52.03 Flushing and Bleeding – Flushing and bleeding is the emergence of excess asphalt and fines onto the finished surface and the significant loss of coarse surface texture, leaving little or no effective cover aggregate visible.

508.52.04 Pock Marks – Pock Marks are indentations in the mat texture, normally less than the thickness of the graded seal layer, evidenced by the absence of effective cover aggregate in a small area. (No photograph available)

508.52.05 Pot Holes – Pot Holes are the loss of all cover aggregate and base material(s) into the underlying asphalt

or gravel base. Pot Holes may develop because of seal and/or base weakness.

508.52.06 Ravelling – Ravelling is a general term for loss of cover aggregate, which can include numerous pock marks.

508.52.07 Streaking – Streaking is a continuous visible absence of effective cover aggregate often seen paralleling a 'ridge' of excess cover aggregate, characterized by the narrowness of its width and the significant extent of its length. Where streaking displays an absence of effective cover aggregate, the condition is unlike ravelling in that there was originally little or no effective aggregate in the area.

508.52.08 Washboard – Washboard is a condition of non-uniform surface texture caused by the buildup of transverse ridges within the graded seal layer itself, and characterized by a very rough, vibratory or chattering ride. (No photograph available)

508.53 End Product Evaluation

508.53.01 Lot Definition – A Lot shall be one lane kilometre of highway sealed**.** Sections of any road/highway remaining which are less than one-half (0.5) kilometres in length, will be included in the previous Lot. Sections remaining which are 0.5 kilometres or greater, but less than one kilometre will receive a Lot designation unique for the road/highway.

508.53.02 Rating Time Criteria – The Ministry Representative will rate each Lot, within thirty (30) days of final brooming.

508.53.03 Performance Rating Criteria – Performance rating will be in accordance with the two tables in DBSS 508 Appendix F: *The Surface Defect Parameters Table* and the *Surface Defects Frequency Definitions,* and the SS 508 photographs.

508.54 Deficiency Repair and Replacement – Any Lot of sealed surface which is found unacceptable shall be remedied in accordance with the (DBSS 508 Appendix F) "Surface Treatment Repair/Replacement Criteria", until the whole project has been repaired or replaced as necessary and accepted by the Ministry Representative.

On double seal applications both the lower and upper lifts must be free of deficiencies prior to being accepted.

The sealing Contractor will be responsible to remedy deficiencies attributed to base, base failures and base work done by others.

For any defects being repaired through re-seal, the repair must cover the entire width of the highway (or to a centerline barrier), unless the Contractor can demonstrate that a compliant repair can be achieved at a narrower width without any longitudinal ridging which may adversely affect driver control or trap water adjacent to the edge of the repair.

508.55 Attention to Potential Safety Hazards – If, in the opinion of the Quality Manager or the Ministry Representative, remedial Work is required to correct a potential safety hazard it shall be undertaken immediately.

PART G - MEASUREMENT AND PAYMENT

508.61 Not Used

508.62 Not Used

508.63 Not Used

508.64 Not Used

508.65 Claims for Vehicular Damage - Without in any way limiting the Contractor's obligations or liabilities herein, during construction and up until final acceptance of the whole project by the Ministry Representative, the Contractor shall be fully responsible for all claims for damages caused by the work and shall address and deal with each claim submitted.

APPENDIX A

Quality Control Tests, to be obtained by the Contractor from the Supplier, for High Float Emulsified Asphalts

Test ASTM VERSIONS	
ASTM D 140	Standard Practice for Sampling Bituminous Materials
ASTM D88	Saybolt Furol Viscosity
ASTM D-113	Ductility Test for Polymer Modified Emulsions
ASTM D-244 (CAN2-16.5-M84)	Residue by Distillation
ASTM D-5 (CAN2-16.5-M84)	Penetration of Asphalt Materials
ASTM D 4957 (CAN2-16.5-M84)	Absolute Viscosity of Asphalts
ASTM D-139 (CAN2-16.5-M84)	Method of Float Tests for Bituminous Materials

APPENDIX B

HIGH FLOAT EMULSIFIED ASPHALT REPORT (H-539)

Project No:_____ **Refinery:** _____

Contract:_____ **Asphalt Grade:**_____

Location:_____ **Batch No:**_____

Sampling Date:_____ **Testing Date:**_____

REQUIREMENTS	RESULTS	SPECIFICATIONS	
		MAXIMUM	MINIMUM
Residue by Distillation % by mass			
Oil Distillation % by volume			
Saybolt Furol Viscosity (SFS) 50°C			
Penetration 5 s/100 g/25° C			
Viscosity, mK or CM (Pa•s) 60° C			
Ductility, at 4°C			
Float Test (s) 60° C			

APPENDIX C

EMULSION USAGE

Project No: _____ **Product:** _____

Date: _____ **Supplier:** _____

Contractor: _____ **Supplier Lading No:** _____

Emulsion Temp.	Start Litres A	Finish Litre B	Litres Used L = A - B	Spray Width(m) D	Spray Length (m) E	Spray Area (m^2) F = D X E	App Rate(L/m^2) L/F	Location

The undersigned certifies that the quantities placed are as indicated in the information above.

Name: (Contract Operator): _____. Date: _____.

Checked by (Ministry Representative): _____. Date: _____.

NB: In the "Location" column give station to station or landmark to landmark inventory kilometre and indicate the lane being sealed (e.g. WBSL = west bound slow lane)

APPENDIX D

DAILY ACTIVITY LOG

1. **General**

Contractor_____Address_____

Superintendent_____Phone_____Fax_____

Project No._____Hwy/Road_____

Project Description_____

Emulsion Manufacturer_____ Emulsion Type_____

Aggregate Source(s)_____Aggregate Class_____

2. **Equipment** (e.g.: type, make and model of chip spreader)

_____
_____
_____
_____
_____

3. **Weather**	**at start-up**	**at shut-down**
Ambient Temperature:	_____	_____
Road Surface Temp.:	_____	_____
General Conditions:	_____	_____

4. **Traffic Control**	**Number/Type**	**Duration (From/to)**
Pilot Vehicles	_____	_____
T.C.P.'s	_____	_____
Signs (specify)	_____	_____

5. **General Remarks (e.g.: weather, dust, traffic volumes, mechanical, materials)**

_____
_____
_____
_____

(Contractor's Representative) (Date)

APPENDIX E

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

1.01 GENERAL

The Contractor's guidelines for preparing, submitting and adhering to a Quality Control Plan are specified in DBSS 101, DBSS 508 Part E, the Contract and this Appendix.

The QC Manager shall have at a minimum 5 years related experience in Quality Management.

1.02 QUALITY CONTROL PLAN

The Plan shall address all elements that affect the quality of the materials, end product seal coat, including but not limited to the following:

- Supply of Asphalt Materials
- Identification of source and proof of quality of aggregates to be supplied
- Aggregates production and its gradation control
- Quality of aggregate components during production
- Stockpile management
- Sealcoat Design(s)
- Distributor and spreader calibration
- Process temperature controls
- Material application rates
- Joints and tapers

1.03 CONTRACTOR'S RECORD OF QUALITY CONTROL TESTING

Test results should be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control during aggregate production should be kept on the form H-295 – Mechanical Analysis of Aggregates, or on a similar gradation analysis form acceptable to the Ministry Representative.

1.04 MATERIAL APPLICATION RATE

The Contractor shall control the Material Application Rate by monitoring the amount of emulsion and aggregate delivered to the road against the area covered by checking the application rates of the distributors and spreader.

1.05 OTHER QUALITY CONTROL PROCEDURES

The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in a Quality Control Plan. Procedures may also be introduced after the start of work as amendments to the Quality Control Plan.

1.06 QUALITY CONTROL TESTING FREQUENCY

Recommended test frequency requirements for Quality Control are described in Table 1 following.

1.07 QUALITY ASSURANCE TESTING FREQUENCY

Quality Assurance test frequency shall be approximately 10% of that described for Quality Control and may vary, in accordance with the Quality Management Plan and the observed effectiveness of Quality Control.

TABLE 1 – TEST FREQUENCY GUIDELINES

	ASTM Test Designation	*Minimum Frequency
Tests Before Aggregate Production	D-75 Sampling Stone, slag, gravel, sand and stone block for use as Highway Materials	When Required
	C-117 Test for Materials Finer than 75 µm (No.200) Sieve in Mineral Aggregates by Washing	1 test (Representative Sample)
	C -142 Test Method for Clay Lumps and Friable Particles in Aggregates	1 test (Representative Sample)
	D 6928 Micro Deval	1 test (Representative Sample)
	D-5821 Fracture Count on Coarse Aggregate	1 test (Representative Sample)
	D-4791 Test Method for Flat Particles, or Flat and Elongated Particles in Coarse Aggregate	1 test (Representative Sample)
Tests During Aggregate Production	C-136, Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates	- Main Stockpile -1 per each 1.5 hours production.
	D-5821, Determining the Percentage of Fractured Particles in Course Aggregate	- Main Stockpile -1 per each 1.5 hours production.
	C-117 – Test for Materials Finer than 75 µm (No 200) Sieve in Mineral Aggregates by Washing	- 1/shift on reduced sample obtained from combined samples from the crusher.
Tests on Emulsified Asphalt Products	D-244, Residue by Distillation	- minimum 1 per batch
	D-244, Saybolt Furol Viscosity	- minimum 1 per batch
	D-5, Penetration of Bituminous Materials	- minimum 1 per batch
	D-139, Float Test	- minimum 1 per batch
	D-4957, Apparent Viscosity on Residue	- minimum 1 per batch
	D-113, Ductility	- minimum 1 per batch on polymer modified emulsions.
Tests on Application of Sealcoat Materials	Emulsion Pan Test	- 3 per distributor/day. Test procedure, variation limits and actions should be outlined in Contractor's Quality Control Plan.
	Aggregate Pan Test	- 2 per spreader/day. Test procedure, variation limits and actions should be outlined in Contractor's Quality Control Plan.

***** These are the recommended minimum frequencies and the contractor is responsible to assess the need to increase testing when required.

APPENDIX F

SURFACE DEFECT AND TREATMENT REPAIR/REPLACEMENT CRITERIA

SURFACE DEFECTS	SEVERITY	RATING PARAMETERS	REPAIR/REPLACEMENT METHOD
Loss of cover aggregate (Ravelling)	Very good	No noticeable aggregate loss	None
	Good	A few pock marks, less than 5 per 0.1 m^2	None
	Fair	Frequent pock marks closely spaced, more than 5 per 0.1 m^2	Reseal of affected area***
	Poor	Extensive pock marks or few surface disintegrations	Reseal of affected area***
	Very poor	Disintegration with potholes	Reseal of affected area***
Potholes	Very good	Few minor potholes, only involves graded aggregate seal	All potholes to be filled with asphaltic patching material at contractor's expense
	Good	Few deep potholes, involves granular base	All potholes to be filled with asphaltic patching material.(see DBSS 508.54)
	Fair	Intermittent potholes	Patch potholes and reseal affected area*** (see DBSS 508.54)
	Poor	Frequent potholes	Patch potholes and reseal affected area*** (see DBSS 508.54)
	Very poor	Extensive or potholes throughout	Patch potholes and reseal affected area*** (see DBSS 508.54)
Flushing / bleeding	Very good	No/very faint colour change in wheel path	None
	Good	Few sections with asphalt on surface	None
	Fair	Intermittent sections with asphalt on surface	Application of sand blotter
	Poor	Frequent sections with asphalt on surface, has wet look or asphalt on surface throughout	Reseal of affected area. Removal of initial surface treatment at Contractor's option***
	Very poor	Wet look with noise like a wet pavement	Reseal of affected area. Removal of initial surface treatment at Contractor's option***
Total failure	Any	Any condition where the asphalt material softens or disintegrates under traffic and aggregate is picked up or "kicked off" by traffic	Remove and dispose of failed surface treatment in its entirety and apply new surface treatment
Streaking (see raveling)	Any	Absence of cover aggregate caused by too little binder - characterized by long, narrow sections of little or no stone cover	Reseal of affected area
Washboard	Any	Any condition of repetitive build-up across the travel lane	Remove and reseal affected area

NOTE:*** Reseal for these defects shall consist of a new graded aggregate surface treatment, single pass application. Area of the reseal repair/replacement shall not be less than one lane width, 10 metres in length. If there is less than 10 metres between two sections in the application pass designated for repair/replacement, the repair/replacement shall be continuous.

APPENDIX F (Continued)

SURFACE DEFECTS FREQUENCY DEFINITIONS

DENSITY OF SURFACE DEFECTS**			
SEVERITY	RAVELLING (% LENGTH)	POTHOLES (NUMBER)	FLUSHING/BLEEDING (% LENGTH)
Very Good	<5%	0 - 1	<5%
Good	5% - 20%	2 - 15	5% - 20%
Fair	21% - 50%	16 - 30	21% - 50%
Poor	51% - 80%	31 - 50	51% - 80%
Very Poor	>80%	>50	>80%

Note: ** Based on number of surface defects or percent of surface affected by defect, per Lot as defined in DBSS 508.53.01.

For illustrative photographs of the surface defects, see SS 508 Appendix F in the Ministry's *Standard Specifications for Highway Construction.*

SECTION 511

COLD MILLING

DESCRIPTION

511.01 Scope - This work consists of preparing a foundation for placement of new surfacing by cold milling the existing pavement surface to the depth and width as indicated in the Contract and the Design.

EQUIPMENT

511.21 Milling Machine - The milling machine may be self-propelled, be equipped with automatic longitudinal and transverse grade control and a profiling ski or boom of a recommended minimum 4 m length, The cutting drum shall be totally enclosed. There shall be an effective means of removing the loosened material from the surface and for preventing dust from escaping into the air.

511.22 "Kill" Switch - All pavement milling machines shall be equipped with a "kill" switch installed or approved by the manufacturer of the milling machine. This "kill" switch shall be automatically actuated whenever the operation of the machine is so impeded that a hazardous situation, such as "kick back", would result were that operation to continue. When so actuated the "kill" switch shall instantly shut down the operation of the milling machine.

511.23 Not Used

CONSTRUCTION

511.31 Not Used

511.32 Pavement Removal - The existing pavement shall be removed to the depth and width as specified in the Contract and the Design. The surface shall be free of longitudinal and transverse irregularities. At all times, during the milling operation, the traveled roadway shall be kept clean of all loose materials.

At the end of the milling shift or at the end of each section to be milled, prior to reopening the road to traffic, the pavement shall be cleaned and swept so that all loose material is removed. Also, the transition from the milled surface to the existing paved surface shall be tapered at a maximum slope of 25:1.

On extensive milling projects no more than one milling lift shall be carried out in any one lane so as to minimize the length of the grade difference between the lanes. At no time, at the end of a shift, shall there be a grade difference that is not at lane dividing lines, centre line. When the milling operation traverses intersecting roads or on and off ramps, the transition from the milled surface to the existing pavement shall be at a maximum slope of 25:1. This transition can be done with the milling machine or by using tar paper letdowns.

If, due to delays between the milling and paving operations, the milled surface starts to pothole or deteriorate, repairs shall be carried out at once using asphalt mix. At no time shall there be a grade difference between lanes of more than 50 mm at the end of a shift. If the milling is carried out to a depth greater than 50 mm, the full width of the driving surface shall be milled or a lift of leveling course shall be constructed to maintain a maximum depth of 50 mm.

The Contractor shall cut drainage channels as required to prevent water from collecting in the milled area.

511.33 Disposal of Materials - The disposal of materials removed under this section which are not recycled and used on the project shall become the property of the Contractor at the point of removal, unless otherwise specified. Materials shall be disposed of outside the limits of the project in a manner that shall meet all environmental requirements. The procurement of a disposal site, if required, shall be the Contractor's responsibility.

511.43 Existing Structures - Care must be taken not to disturb or damage any structures or devices such as manholes, catch basins, valves, boxes and other utilities. Damage to visible, referenced, or plan-indicated manholes, catch basins, valves, valve covers and concrete/asphalt curb, or any other infrastructure shall be repaired. All metal and concrete faces must be cleaned of old pavement and painted with primer, prior to repaving.

SECTION 514

HOT IN-PLACE RECYCLE

DESCRIPTION

514.01 Scope - This Specification describes the equipment, work practice and materials required for carrying out Hot In-Place Recycle operations.

MATERIALS

514.11 Rejuvenating Agent-Requirements - The Contractor shall supply an approved rejuvenating agent that is listed on the Ministry's Recognized Products List, which may be found on-line at:

http://www.th.gov.bc.ca/publications/eng_publications/geotech/rpb.htm

514.12 Asphalt Mix - When the Contract requires the addition of virgin mix to be incorporated into the recycled asphalt mix, it shall meet the requirements as shown in the Contract.

EQUIPMENT

514.21 General - All equipment and instrumentation shall be capable of performing the work in accordance with these specifications.

514.22 Processing Equipment - The equipment shall have the capability to process the existing pavement to a minimum depth of 50 mm. The equipment shall be capable of heating the existing pavement to a width 100 to 200 mm wider than the width to be processed and to a temperature, behind the screed, of not less than 110°C. The equipment shall be regulated so that excessive heating and burning of the existing asphalt cement does not occur. This will generally require the existing surface to be radiantly heated and no open flame will be permitted.

The scarifying tines or cutting drum(s) shall mill to the specified cross-section and shall be set so as to produce a fully homogenous recycled mixture free of lumps.

The equipment shall have an activated heated screed complete with augers and strike-off device, capable of distributing and placing the reprocessed mix to the full width of the pavement being recycled.

Rejuvenating agents that contain water will not be permitted.

The hot recycling unit shall be equipped and capable to add and homogeneously blend any specified proportion of virgin Asphalt Mix into the reclaimed material and shall be equipped with a means to introduce a rejuvenating agent homogeneously into the reclaimed mixture.

The Contractor's compaction equipment shall conform to

TABLE 514-A SPECIFICATIONS

BCH Test	ASTM Test	TITLE OF TEST	SPECIFICATION
II-1	D-2726	Bulk Relative Density of Compacted Bituminous Mixtures using Saturated Surface Dry Method	
II-2	D-1188	Bulk Relative Density of Compacted Mixes Using Paraffin Coated Method	
II-3	D-1559	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	
III-4	D-2170	Kinematic Viscosity @ 60°C	200-500 mm²/sec
III-7	D-70	Specific Gravity	0.98-1.02
III-11	D-92	Flash Point by Cleveland Open Cup	204°C Min.
	D-1160 (10 mm Hg)	Volatility - IPB - 2%v. - 6%v.	 150°C min. 190°C min. 210°C min.
	D-2007	Saturates, %w	28 max.
	D-2006-70	Asphaltenes, %w	1.5 max.
	D-2006-70	Chemical Composition $(PC+A_1)/(S+A_2)$	0.4-0.8

DBSS 501.36 in its entirety.

Individual pre-heating and reprocessing units shall be equipped in such a way that the operator and other workers are not exposed to any hazardous fumes or gases produced from the equipment operation, the heated pavement or from the addition of the rejuvenating agent, in accordance with all applicable regulations.

514.23 Rejuvenating Agent Equipment - To accurately proportion the rejuvenating agent into the scarified mixture the equipment shall comply with the following requirements:

a) positive feed and shut off of the rejuvenating agent, governed by the movement of the reprocessing machine

b) control of the quantity of rejuvenating agent to ± 0.05 ℓ/m² of surface reprocessed with an agent application range of 0.0 ℓ/m² to 1.0 ℓ/m²

c) measurement of the rejuvenating agent applied by means of a metering device, which is capable of recording accumulated litres to the accuracy of "± 2%, and which is fully visible to both the operator and the Ministry Representative

d) mechanically or electronically controlled application of the rejuvenating agent relative to and variable with the processing rate of the equipment

e) a mechanical or electronic metering device capable of supplying an accurate indication of the quantity of rejuvenating agent being applied per square metre

f) a mechanical, electronic or manual method of measuring the quantity of rejuvenating agent in the tank

g) heating of the agent to within ± 5°C of the application temperature established by the manufacturer.

CONSTRUCTION

514.31 Preliminary The existing paved surface shall be cleaned to avoid contamination of the recycled pavement. It shall be swept back at least 20 cm wider than the area to be processed.

Prior to proceeding with the road work, the Contractor shall ensure that all catch basins and manholes, which are within the operating area of the heating units, are clear of any flammable liquids or otherwise hazardous fumes/gases. **514.32 Recycling -** The existing highway pavement shall be heated, planed, reprocessed, with or without the addition of a rejuvenating agent and with or without the addition of virgin mix as specified, remixed, replaced and compacted using a heater scarification or hot milling process and rollers.

The recycled material shall not be heated over an average of 150°C in order to avoid excessive oxidation and hardening of the recycled Asphalt Cement.

At the locations where the recycling operations begin and end, the Contractor shall ensure that the transition areas between the processed and unprocessed surface is smooth. If any bumps or irregularities occur resulting from the reprocessing operation, the Contractor shall repair these areas with fresh asphalt mix and/or additional reprocessing.

At all times the Contractor shall ensure that the appropriate cross slopes for safety and positive pavement drainage are maintained.

The Contractor shall ensure that the longitudinal edges of the recycled pavement are blended to conform in elevation with the adjacent pavement unless this surface is scheduled to be recycled.

Generally, the outer edge of each lane will be tapered to a 12:1 slope.

Excess materials shall be removed and deposited in a location such that they can be reincorporated into the recycled mixture. At no time shall excess material be cast across the newly processed surface or incorporated into the granular shoulders.

514.33 Compaction and Finishing - Immediately behind the reprocessing operation, the Contractor shall supply roller compaction to obtain a minimum of 97% laboratory density. The laboratory density will be the specimen density, (ASTM Test D 2726 or D 1188) that results from heating a representative sample of the existing pavement, or rejuvenated and blended mixture, as applicable, to a temperature of 130°C and producing a compacted laboratory specimen using the procedure for 75 blow Marshall method. (Sample preparation shall be in accordance with applicable portions of ASTM Test D 1559).

The mix for compaction testing shall be obtained behind the screed of the laydown machine, and the roadway shall be marked at that location for subsequent drilling of the core. The core shall be cut at the specified heater recycling thickness prior to obtaining its density.

The reprocessed mat shall have a uniform appearance. Any deficiencies shall be corrected to the satisfaction of the Quality Manager and the Ministry Representative.

The completed Hot In-Place Recycled surface shall conform to all requirements of DBSS 501.10.

SECTION 515

END PRODUCT SPECIFICATION FOR HOT-IN-PLACE RECYCLED ASPHALT PAVEMENT

PART A - GENERAL

515.01 General – This Section describes the materials, equipment, professional standards and end product requirements for the construction of Hot In Place Recycled Asphalt Pavement.

End Product Specifications contain the acceptance and payment criteria based on the results of specified sampling and testing. Payment is subject to adjustments upward and downward in accordance to the provisions provided in End Product Specifications (EPS).

515.02 General Description Of Work - The Contractor will be responsible for the following work associated with the construction of hot in place asphalt pavement by EPS:

- Preparing a Quality Control Plan for evaluation before commencing the Work and providing at the production site a testing facility to provide the data needed to implement that plan.
- Supplying, screening, crushing, processing and improving aggregate to produce virgin asphalt mix aggregate.
- Supplying and delivering asphalt cement meeting the requirements of DB SS 952.
- Supplying and delivering rejuvenating agents meeting the requirements of DB SS 514.11 and DB SS 514.23.
- Heating the virgin asphalt mix aggregate and mixing it with asphalt cement to produce asphalt admix that meets the Job Mix Formula.
- Hauling, and adding the asphalt admix into the recycling process.
- Recycling the existing highway pavement by heating, milling, reprocessing, with or without the addition of rejuvenating agents or admix (virgin mix) as specified, re-mixing, replacing and compacting using a multi-stage, hot milling process and rollers.

The Contractor shall provide an end product conforming in quality and accuracy of detail to the dimensional and tolerance requirements of the contract. Where no tolerances are specified, the standard of workmanship shall be in accordance with normally accepted good practice and those identified in DB SS 514. Payment is subject to adjustments based on quality audit tests performed by the Ministry Representative.

The Contract may provide information pertaining to the gradation of the admix, asphalt content, addition rates for rejuvenating agents and the percentage of admix to be incorporated.

515.03 Definitions Relevant To End Product Specification

515.03.01 End Product Specification (EPS) - A specification whereby the Contractor is entirely responsible for quality control of the construction processes, and whereby the Ministry performs the specified quality audit sampling and testing of the end product for the purpose of determining acceptance/rejection and payment adjustments.

515.03.02 Job Mix Formula - The Job Mix Formula, supplied by the Contractor, establishes the virgin aggregate proportions, gradation, the asphalt content and type of asphalt to be used for production of asphalt admix. The Job Mix Formula is based on an asphalt mix design, in accordance with DB SS 515.05.06, or on a variation in accordance with DB SS 515.05.07.

515.03.03 Asphalt Admix - The Asphalt Admix is virgin asphalt mix added to the recycled mix. The Asphalt Admix is project specific designed and is blended homogeneously within the recycling process. The Asphalt Admix becomes an integral component of the recycled pavement structure.

515.03.04 Asphalt Content - Asphalt Content means the quantity of asphalt cement in the virgin admix expressed as a percentage by weight of the total dry aggregate in the mix.

Design Asphalt Content - The asphalt content upon which the Job Mix Formula is initially established.

Approved Asphalt Content - The design asphalt content or subsequent adjustment to it, embodied in a Job Mix Formula or revised Job Mix Formula.

Actual Asphalt Content - Actual asphalt content is the amount of asphalt cement in the admix as determined by the Contractor's Quality Control Testing.

515.03.05 Smoothness - Smoothness is a measure of the longitudinal profile of the pavement surface. The unit for measurement is the International Roughness Index (IRI).

515.03.06 Sample Mean - The Sample Mean is the arithmetic mean of a set of test results constituting the sample.

515.03.07 Stratified Random Sample - A Stratified Random Sample is a set of test measurements taken one

each from a number of separate (stratified) areas or Sub-Lots within a Lot in an unbiased way.

515.03.08 Lot - A Lot is a portion of the work being considered for acceptance and for the determination of payment adjustments. A Lot is defined as follows.

For the application of the contract requirements for:

- Density,

A Lot shall be one day's scheduled production of at least 7 hours of pavement recycling where no changes have occurred to criteria such as but not limited to:

- approved Job Mix Formula,
- the specific thickness to be processed,
- the required materials addition rate,
- change in the ratio of the components making up the Hot In Place Recycled Pavement.

A change in any above criteria may require a new Lot designation.

One day's production of less than 7 hours will be dealt with as follows:

- the material will be added to the next Lot that has the same criteria, as described above, except that
- if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot,

A Lot shall be no more than two days total production even if the above criteria have not changed or been met.

For application of the contract requirements for:

- Smoothness,

A Lot shall be one kilometre length of recycled pavement for each driving lane.

515.03.09 Sub-Lots for Density and Smoothness - For the application of the contract requirements for:

- Density, each Lot shall be divided into three equal Sub-Lots.
- Smoothness, each Lot shall be divided into 100 metre Sub-Lots.

515.03.10 Driving Lane - A driving lane shall mean a single lane in any area of the pavement other than a shoulder or a barrier flare.

515.03.11 Not Used

515.03.12 Hot In Place Recycled Pavement - Hot In Place Recycled Pavement shall be the result of the process where recycling, the addition of admix and/or rejuvenating agents, the spreading, compacting and finishing of the asphalt pavement have taken place to form a quality finished product. The Hot In Place Recycled Pavement may be comprised of the following components: existing asphalt pavement, admix, and/or rejuvenating agents.

515.04 Quality Control

515.04.01 General - Quality Control, by the Contractor, is the sum total of activities by the Contractor to ensure that a product meets contract specification requirements. Quality Control includes material handling and construction procedures, calibration and maintenance of equipment, production process control and any sampling, testing and inspection that is done for these purposes.

The Contractor is responsible for all Quality Control under this specification. The Ministry Representative will audit and monitor the Contractor's operations and the implementation of the Contractor's Quality Control Plan.

The Ministry will not take samples for quality control testing and will in no manner assist in any degree or in any aspect of the Contractor's operation in the production of asphalt pavement, beginning from the production of aggregate through the compaction of asphalt mix. The only exception shall be in accordance with the provisions of DB SS 515.05.07, where the Ministry shall have the ability to adjust the ratio of the components of the Hot In Place Recycled Pavement based upon the Contractor's Quality Control Results and visual inspections.

515.04.02 Quality Control Plan - The Contractor shall prepare a detailed, written Quality Control Plan, based on the guidelines as shown in Appendix 1 and functions integrally with any other Quality Management provisions of the Contract. The Contractor's Quality Control Plan shall be submitted to the Ministry Representative for information before starting the production of paving aggregate

The Contractor's Quality Control Plan shall include a description and schedule of the intended sampling, testing and reporting. The Quality Control Plan must also include a detailed description of the means by which the Contractor shall use the quality control test results to ensure that the asphalt materials, aggregate, mix production, recycling and pavement compaction processes will be controlled to keep the end product within the specified limits. The Quality Control Plan must clearly show the flow of information from the quality control laboratory to the individuals who shall make the actual adjustments to the processes and equipment to this control. The plan will show time allowance for each step, the

names and positions of all the people involved, and a clear description of the responsibilities of each.

515.04.03 Quality Control Testing and Inspection - The Contractor shall provide and maintain equipment and qualified personnel to perform all laboratory testing, field testing and inspection necessary to determine and monitor the characteristics and properties of all the materials produced and incorporated into the work. They shall also monitor the workmanship of the final product in accordance with the Quality Control Plan as most recently submitted.

The Contractor shall use a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified, registered member of the Applied Science Technologists and Technicians of British Columbia. This person shall be designated as the Quality Control Manager for the purposes of these specifications. The Quality Control Manager shall be responsible for preparation and sign off of the Quality Control Plan, shall be responsible for all Quality Control testing and inspections and shall be responsible for the signing of all Quality Control testing and inspection records and submissions to the Ministry.

The Contractor shall provide a testing facility(s) that meets the requirements necessary to carry out all the test procedures listed within this Section. The facility(s) must have the equipment specified under the appropriate test designation to perform the tests.

All equipment and the laboratory shall be well maintained and in good working condition. All testing equipment shall be calibrated and evidence of the calibration shall be provided to the Quality Manager, and, upon request, to the Ministry Representative.

515.04.04 Quality Control Records - The results from quality control testing shall be reported on test logs and plotted on charts immediately after each test is completed. These test results, reports and charts shall be available, to the Ministry Representative upon request.

For the purposes of confirming delivery of asphalt mix to the road and the calculation of material application rates, the Contractor shall prepare and provide to the Ministry Representative, a copy of the Road Checker's Summary.

515.04.05 Final Quality Control Testing Reports - The Contractor shall retain and provide the Ministry Representative on request :

- a summary of all virgin aggregate quality control test results;
- copies of all quality control test results for asphalt admix properties, Hot In Place Recycled Pavement and compaction; and
- copies of all quality control charts.

515.04.06 Minimum Acceptable Construction Practices - Good construction practices shall be considered as standard construction procedures to be followed and shall include but not be limited to the procedures described below.

Stockpiles – Stockpiles of different types of material shall be located and constructed in such a manner as to prevent intermingling of the types and segregation of material.

Haul Vehicles – Lubrication of the truck boxes with diesel fuel will not be permitted. All vehicles shall have adequately insulated truck boxes and shall be equipped with an insulating tarpaulin of such size as to completely protect the asphalt mix.

Existing Paved Surfaces – Existing paved surfaces shall be clean to avoid contamination of the recycled pavement. The existing surfaces shall be swept back at least 20cm wider than the area to be processed.

Existing Utility Structures – The Contractor shall ensure all catch basins and manholes, which are within the operating area of the heating units are clear of any flammable liquids or otherwise hazardous fumes/gases.

Heating of Recycled Material – The recycled material shall not be heated over an average of 150°C in order to avoid excessive oxidation and hardening of the recycled Asphalt Cement.

Minimum Admix Temperature – Based on the most recent Temperature Viscosity Curve supplied by the Asphalt Cement Supplier. This information will be used to set minimum and maximum mixing temperatures.

TABLE 515-A-1 STANDARD SAMPLING AND TESTING PROCEDURES FOR VIRGIN AGGREGATES

All virgin aggregate tests shall use the following sieve sizes:
37.5mm, 25mm, 19mm, 16mm, 12.5mm, 9.5mm, 4.75mm, 2.36mm, 1.18mm, 0.600mm, 0.300mm, 0.150mm, and 0.075mm.

Item	Procedure	Test Reference
1	Sieve Analysis of Fine And Coarse Aggregates	ASTM C-136
2	Materials Finer Than 0.075 mm (No. 200) Sieve in Mineral Aggregate by Washing	ASTM C 117
3	Specific Gravity and Absorption of Coarse Aggregate	ASTM C-127
4	Specific Gravity and Absorption of Fine Aggregate	ASTM C-128
5	Determining the Percentage of Fractured Particles in Coarse Aggregate	ASTM D-5821
6	Total Moisture Content of Aggregate by Drying	ASTM C-566

TABLE 515-A-2 STANDARD SAMPLING AND TESTING PROCEDURES FOR ASPHALT CEMENT

Item	Procedure	Test Reference.
1	Penetration of Bituminous Materials	ASTM D-5
2	Viscosity of Asphalt by Vacuum Capillary Viscometer	ASTM D-2171

TABLE 515-A-3 STANDARD SAMPLING AND TESTING PROCEDURES FOR VIRGIN ASPHALT ADMIX AND HOT IN PLACE RECYCLED PAVEMENT

Item	Procedure	*Test Reference*
1	Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures	ASTM D-2726
2	Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures	ASTM D-2041
3	Sampling Compacted Bituminous Mixtures for Laboratory Testing	ASTM D-5361
4	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	ASTM D-1559
5	Determining Asphalt Content by Ignition Methods for Admix.	ASTM D-6307

Longitudinal Joints – Longitudinal joints in the top lift of asphalt pavement will generally only be permitted where lane dividing lines are to be painted.

Longitudinal Edges – Are to be blended to conform in elevation with the adjacent pavement unless this surface is scheduled to be recycled.

Minimum Temperature – Minimum Temperature behind the paver screed of the recycled material shall not be less than 110°C.

Metering of Rejuvenating Agents – To accurately proportion the rejuvenating agent into the hot milled material, the equipment shall control the quantity of the rejuvenating agent to ±0.05 ℓ/m^2 of surfaced reprocessed with an agent application range of 0.0 ℓ/m^2 to 1.0 ℓ/m^2. The measurement of the rejuvenating agent applied by means of a metering device shall be capable of recording accumulated litres to the accuracy of ±2%, and be fully visible to both the operator and quality management personnel.

Outer Edges – The outer edge of each lane shall generally be tapered to a 12:1 slope.

Recycling Adjoining Mats – When recycling an adjoining parallel mat there shall be at least a 50-mm overlap onto the previously recycled surface. This overlap shall be properly "raked" to form a homogeneous bond between the two mats.

Raking – Any material that is raked from the joint shall not be placed on the new mat or placed on or in front of the paver. This material may be introduced at a point of the operation where the material shall be reheated and remixed with the Hot In Place Recycled Pavement.

Discontinued Paving - At locations where reprocessing operations begin and end, the Contractor shall ensure that the transition between the processed and unprocessed surface is smooth and without irregularities. If any irregularities occur resulting from the reprocessing operation, the Contractor shall repair these areas with fresh asphalt mix and/or additional reprocessing.

Pavement Drainage – The Contractor shall at all times ensure that the appropriate cross slopes for safety and positive pavement drainage are maintained.

Rollers – Rollers shall normally operate with the drive wheel nearest the paver and at a speed not in excess of 8 km/h. They shall not be allowed to park on the mat prior to complete cooling.

TABLE 515-A-4 TEST SAMPLE SOURCE LOCATIONS

Test	Sample Source:
Density	150mm diameter. road cores
Smoothness	Centre of Lane

515.05 Supply Of Virgin Aggregates, Asphalt Materials, Asphalt Admix Design And Job Mix Formula

515.05.01 Work in Ministry Pits or Quarries - When operating in a Ministry pit or quarry, the Contractor shall comply with all provisions of DB SS 145.26.

515.05.02 Supply of Virgin Aggregates, Aggregate Production and Characteristics - The Contractor shall not produce paving aggregate until the Contractor has received written notification that the Quality Control Plan is in accordance with DB SS 515.04.02, and has in place testing facilities for aggregate production that are in accordance with the Quality Control Plan.

Where the Ministry has available any test result information on the properties shown in Table 515-B, for a Ministry pit or other source, the Ministry will upon request, provide that information to the Contractor. Otherwise, sampling and testing to determine and demonstrate the compliance of paving aggregate with the requirements of this Section shall be the responsibility of the Contractor.

Paving aggregates shall meet the following requirements:

a) Virgin Coarse Aggregates;

i) shall be all mineral matter retained on the sieve designated in the test procedures for each individual test.

ii) shall consist of crushed stone, crushed gravel, or combination thereof, or materials naturally occurring in a fractured condition, or materials naturally occurring of highly angular nature or rough texture.

iii) shall be free from coating of clay, silt or other deleterious material, and shall meet the requirements listed in Table 515-B.

b) Virgin Fine Aggregate

i) shall be clean, tough, durable, moderately sharp, and free from coatings of clay, silt, or other deleterious material, and shall contain no clay balls or other aggregations of fine material.

ii) shall have a sand equivalent of not less than 40 when tested in accordance with ASTM Test D-2419

iii) shall have a minimum value of 45 when tested according the AASHTO Test T 304, Method "A", - Uncompacted Void Content of Fine Aggregate when determining Fine Aggregate Angularity.

c) Virgin Mineral Filler and Mineral Dust

i) Mineral filler shall consist of all matter passing the 0.600-mm sieve and mineral dust shall consist of all mineral matter passing the 0.075-mm sieve.

ii) Mineral filler and mineral dust shall be free from organic matter.

iii) Mineral filler shall be non-plastic when tested in accordance with ASTM Test D-4318.

515.05.03 Supply of Asphalt Cement and Rejuvenating Agents - The Contractor shall supply the types and grades of asphalt cement and rejuvenating agents as specified in the Contract and the Design. The supply of these materials includes, but is not limited to, ordering,

scheduling delivery of, supply of Temperature Viscosity Curve information, receiving, handling, storing, heating, blending, sampling, and testing of the materials and other related work.

The Contractor shall retain and supply the Ministry Representative on request with copies of the supplier's weigh-bill and records of all asphalt and rejuvenating materials.

515.05.04 Responsibility for Asphalt Admix Design Preparation of the asphalt admix design is the responsibility of the Contractor.

515.05.05 Evaluation of Asphalt Admix Materials - The Contractor's Quality Manager will evaluate the Asphalt Admix materials.

515.05.06 Samples Required for Asphalt Admix Evaluation - At the discretion of the Ministry Representative, the Ministry may request samples of the Admix materials and Quality Control results and documentation for evaluation. Shipping costs for samples delivered to the Ministry Representative are the responsibility of the Contractor.

515.05.07 Field Adjustment of the Job Mix Formula and/or Recycled Pavement - During construction, the Contractor may require field adjustments to the job mix formula and/or recycled pavement. A field adjustment to the Job Mix Formula is defined as a change in the asphalt cement content of the admix, aggregate gradation, rejuvenating agent and/or proportioning of various aggregate sizes.

PART B - PAYMENT ADJUSTMENTS

515.11 Density

515.11.01 Coring - The Contractor shall be responsible for providing all core samples for quality assurance and payment adjustment purposes. The location of the cores shall be randomly selected and provided to the Contractor by the Ministry Representative and the cores shall be taken within a 300 radius of that location, unless otherwise agreed by the Ministry Representative. The Contractor shall provide 150 mm diameter cores for these purposes. The minimum core thickness shall be 37 mm. The Contractor shall prepare the cores prior to submission by removing all material not representative of the pavement lift to be tested. The Contractor shall deliver these cores to the Ministry Representative on site, within 24 hours of being provided the locations for the coring.

In the case of a core sample location falling in an obvious non representative area, such as a previously patched area, at the sole discretion of the Ministry Representative, a new random core location will be chosen for that sample.

All costs associated with obtaining the cores, including the filling and compaction of the core holes are considered incidental to the contract and are the responsibility of the Contractor.

515.11.02 Percent Density - One random core sample will be obtained from each Sub-Lot and tested. The test results for the three Sub-Lots will be averaged to determine the percent density for the Lot.

$$\text{Hot-In-Place \% Density} = \frac{\text{In Place Density of Sample}}{\text{Maximum Theoretical Density (ASTM D-2041)}} \times 100$$

The Maximum Theoretical Density for the lot will be determined by combining the Sub-Lot cores. Prior to testing to determine the maximum theoretical density, the cores should be prepared in accordance with the procedure outlined in Appendix 3.

515.11.03 Payment Adjustments - The payment adjustment for percent density will be the amount shown in Table 515 C for the Sample Mean of the test results for the Lot.

TABLE 515-B – REQUIREMENTS FOR VIRGIN COARSE AGGREGATES

Test Reference	Requirements	
ASTM C-127	Maximum Water Absorption: % by Mass	2
ASTM C-88	Soundness of Aggregate Maximum Loss After 5 Cycles	20
ASTM C-142	Maximum % by Mass of Clay Lumps and Friable Particles	1.5
ASTM D 6928–	Standard Test Method for Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	17 minimum
ASTM D-5821	2 Fractured Faces: Minimum % by Mass Retained on the 4.75 mm sieve	85

515.11.04 Rejection Limit - The rejection limit for percent density is the limiting value of the Sample Mean as shown in Table 515 C.

If the test result for density of a Sub-lot is outside the acceptance limits, the Sub-lot is rejected automatically regardless of the values of other acceptance parameters. To minimize the cost of rejection to the Contractor, the Contractor shall isolate the area of low density within the Sub-lot and perform the necessary corrective measures to ensure specifications are met. The limits of the low density area must be verified and approved by the Quality Manager before remedial work proceeds.

TABLE 515 C PAYMENT ADJUSTMENTS FOR DENSITY

% MAXIMUM THEORETICAL DENSITY LOT AVERAGE	PAYMENT ADJUSTMENT ($ PER m^2) FOR DESIGN LOT
95.6 Plus	+ $0.25
95.0 – 95.5	+ $0.15
94.0 – 94.9	+ $0.10
93.0 – 93.9	0.00
92.6 – 92.9	- $0.10
92.0 – 92.5	- $0.25
91.9 or less	Reject

TABLE 515-D – LOT ASSESSMENT AND PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, the following payment adjustments shall apply to each Lot.

Lot IRI (m/km)	Payment Adjustment
≤ 0.80	+$2,000
0.81 to 1.0	+$1000
1.01 to 1.10	+$500
1.11 to1.40	0
1.41 to 1.6	-$250
1.61 to 1.7	-$500
1.71 to 1.99	-$1,000
≥ 2.00	Reject and corrective work required

515.11.05 Payment for Rejected Work Made Acceptable - The payment adjustment for density will be based on testing of the reprocessed, replaced, or overlaid material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work.

515.12 Smoothness

515.12.01 Determination of Pavement Smoothness - The finished pavement surface shall be tested by the Ministry using a Class I precision rolling profile measuring instrument, to determine the longitudinal profile and compute the International Roughness Index (IRI) in each driving lane. Profiles shall be measured and the IRI calculated in the centre of the lane for each Sub-Lot. IRI values will be recorded to a precision of 0.01 m/km for each Sub-Lot. The Lot IRI value is the average of the IRI values calculated for the Sub-Lots within the Lot.

For any Sub-Lot between 50 m and 100 m in length, the IRI value shall be considered representative of a complete Sub-Lot. For any Sub-Lot less than 50 m in length, the IRI value will be combined with the proceeding Sub-Lot IRI value.

The profile shall be measured over the entire length of the pavement exclusive of structures and shoulder areas. Acceleration, deceleration and turning lanes are considered part of the driving lanes and shall be tested in accordance with this provision. For the measuring process, the Contractor shall provide a chalk guide line in the centre of the lane immediately prior to measurement.

515.12.02 Auxiliary Lanes - For smoothness testing, sections of the driving lanes that do not fall within the continuous through lanes, such as acceleration lanes, deceleration lanes and turning lanes, and lanes which are less than 1 km in length, shall be treated as follows. The ratio of the section length to the standard Lot length of 1 km shall be determined and the payment adjustment shall be pro-rated on this basis as in the following example:

Length of segment, i.e. 565 metres = *0.565 times the Standard Lot Length of 1000m.*

Hence the applicable payment adjustment is 0.565 times the payment adjustment for a 1-km Lot as determined from Table 515-D.

515.12.03 Acceptance Limits - The acceptance limit for smoothness is the limiting value as shown in Table 515-D, beyond which corrective work is required. Payment adjustments are shown in Table 515-D. The International Roughness Index (IRI) value, calculated for each Sub-Lot, will be used to determine if the Lot will be accepted, and if

so whether it will be subject to any payment adjustment.

515.12.04 Payment Adjustments for Full or Increased Payment - Acceptance of any Lot at full or increased payment will occur if it contains no obvious defects as per DB SS 515.22 and in the case of top lift pavement only, the pavement smoothness, as reported by the International Roughness Index (IRI) meets the following requirements:

- All Sub-Lots shall have an IRI value ≤ 2.0

515.12.05 Rejection Limit - If a Sub-Lot has an IRI ≥2.0, the Sub-Lot is rejected.

515.12.06 Remedial Work - If the test results on a Sub-Lot of pavement indicate a payment reduction or rejection because of smoothness, the Contractor may propose remedial work to improve the smoothness. Such proposals are subject to the approval of the Quality Manager and the Ministry Representative, but such approval does not imply that the proposed remedy will be successful, and does not reduce the Contractor's responsibility for meeting the acceptance requirements. Reprocessing may be acceptable, but cold milling and repaving may be required. Only one attempt may be made to improve smoothness, and this must be completed within ten (10) calendar days from the time the Contractor receives notification from the Ministry Representative of the original smoothness test results for that Sub-Lot.

Following any attempt to improve the smoothness of a Sub-Lot or Sub-Lots, the Ministry Representative will retest the Sub-Lot or Sub-Lots, and the new results will replace the previous data for the purposes of determining acceptance and payment.

515.12.07 Smoothness Deficiency Repairs - Smoothness deficiencies (bumps and dips) less than 8 mm over 3 m will not have a fix or remedial work requirement. Individual Smoothness deficiencies between 8 mm and 12 mm over 3 m will result in a $200.00 penalty per deficiency and Smoothness deficiencies over 12 mm over 3 m will require remedial work. (see DB SS 515.22).

PART C - ACCEPTANCE AND REJECTION

515.21 Acceptance At Reduced Or Adjusted Payment - Acceptance of any Lot at reduced payment will occur if it contains no obvious defects as per DB SS 515.22, and if:

- the test results for acceptance parameters are such that the Lot or Sub-Lot meets the requirements for acceptance at a reduced payment; and
- the Lot or Sub-Lot is approved in respect of all other requirements; and if
- the Contractor has not notified the Ministry Representative in writing that it will exercise its option to either repair or remove and replace the work, at its own cost, with work meeting the requirements for acceptance at full or increased payment.

515.22 Rejection For Workmanship Defects - The finished surface of any lift shall have a uniform "closed" texture and be free of visible signs of poor workmanship. Any obvious defects as determined by the Quality Manager or Ministry Representative such as, but not limited to the following, will be cause for automatic rejection of asphalt pavement regardless of the values of any other acceptance parameter:

- individual bumps and dips that exceed 12 mm over 3 m.;
- areas of excess or insufficient asphalt;
- improper matching of longitudinal and transverse joints;
- roller marks;
- tire marks; or
- cracking or tearing.

When asphalt pavement is rejected by reason of obvious defects, the minimum area of rejection will be the actual length of the defect for the full width of the driving lane in which the defect exists.

Rejected work shall be promptly repaired, remedied, or removed and replaced in a manner acceptable to the Quality Manager and Ministry Representative. The Contractor shall be responsible for all costs including materials.

No payment will be made for work in any Lot or Sub-Lot, which has been rejected, until the defects have been remedied.

515.23 Appeal Testing

515.23.01 Density - The Contractor may appeal the results of acceptance testing for density for any lot only once. Appeals will only be considered if quality control results can be presented to support the appeal.

Quality control test results for density which are provided to the Ministry Representative subsequent to the Contractors receipt of the quality assurance test results for that Lot will not be considered (when evaluating evidence) for an appeal. The appeal shall be for all tests within the Lots, and there will be no appeal allowed for single tests within the Lot.

The following procedures will apply for the appeal:

- The Ministry Representative will arrange for an independent testing laboratory to perform the

appeal testing. The personnel employed or testing laboratory retained by the Contractor for quality control testing on the project will not be used for appeal testing.

- The original quality audit results will not be considered in the appeal. The Contractor will sample the pavement at locations randomly re-selected by the Ministry within three site occupancy days following receipt of the appeal such that total of (5) new core specimens are extracted throughout the Lot. The Ministry Representative will observe the sampling process. The procedures used to determine the test results shall be consistent with the original acceptance testing methods.

515.23.02 Smoothness - The Contractor may appeal smoothness acceptance test results of any rejected or penalised Lot, once. The appeal shall be in writing and submitted within the next three site occupancy days following receipt of the test results.

Any attempt to improve smoothness on the appealed Sub-Lot after the Ministry Representative has tested the Lot shall void the appeal and the original test results will apply.

The Ministry Representative will perform, and the Contractor will be given the opportunity to witness, the appeal testing and the new results will be binding on the Contractor and the Ministry.

515.23.03 Application of Appeal Testing Results - For density appeals, the original test will be discarded. A new Sample Mean for the five new test results will be determined and used for acceptance and payment adjustment.

The new values, thus determined, in all cases, will be binding on the Contractor and the Ministry.

515.23.04 Payment for Appeal Testing - If the new results indicate a change in the payment adjustment in the Contractor's favour, then sampling and testing costs incurred during the appeal procedures for that Lot will be borne by the Ministry.

If the new results verify that any payment reduction or rejection remains valid for that Lot, then the costs of sampling and testing (plus 10% mark-up) incurred during the appeal procedure will be charged to the Contractor.

515.23.05 Time Limits for Appeals of Test Results - All appeals shall be in writing and submitted within the next three days of receipt of the test results.

515.24 Cold Milling And Re-Paving As A Corrective Measure - If cold milling and re-paving is used as a corrective measure on a defective Lot or Sub-Lot, the thickness will be subject to the approval of the Ministry Representative, but shall not be less than 40mm. In all other respects, the re-paving will be subject to the same specifications as the pavement being replaced.

Whether the cold milling and re-paving is applied as a corrective measure, acceptability and payment adjustment will be determined as follows:

- acceptability, and eligibility for either positive or negative payment adjustment, will be determined entirely on the results of testing and observations conducted on the re-paving, regardless of test results that have been obtained on the hot-in-place recycled pavement;
- the quantity, to which any payment adjustment is to be applied, will be derived from the square meters of recycled pavement affected by the re-paving.

PART D - PAYMENT

515.31 General - Applicable payment adjustments (Additions or subtractions as applicable) shall be applied in accordance with Part B of this Section

The first 4,000 m2 of hot in place asphalt pavement recycling production will not be subject to bonus/penalty payment adjustments but will be required to meet minimum Specification requirements for quality and workmanship.

515.31.01 Not Used

515.31.02 Not Used

515.31.03 Not Used

515.31.04 Not Used

515.31.05 Payment for Acceptable Work - The following end product properties of recycled asphalt pavement will be measured for acceptance:

- Density
- Smoothness

515.31.06 Payment for Rejected Work Made Acceptable - When defects have been remedied in Lots or Sub-Lots which had been rejected, payment for the original quantity of material in those Lots or Sub-Lots will be made subject to payment adjustments and penalty assessments and subject to DB SS 515.24.